

## CHAUTAUQUA COUNTY AP, NY, USA

WMO: 725235

Lat: 42.150N

Lon: 79.250W

Elev: 525

StdP: 95.17

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 04720

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -17.7      | -15.3 | -22.1                         | 0.6 | -16.7 | -19.0 | 0.7 | -13.9 | 11.4                  | -3.7 | 10.4 | -4.0 | 3.4                   | 240  | 0.551 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 9.1                    | 28.0            | 20.9 | 27.3 | 20.3 | 26.1 | 19.5 | 22.4                | 26.5 | 21.5 | 25.3 | 20.7 | 24.3 | 4.2                  | 230  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 21.2                            | 16.9 | 25.0 | 20.2 | 15.9 | 23.8 | 19.1 | 14.8 | 22.9 | 68.5          | 26.3 | 65.1 | 25.3 | 62.0 | 24.6 | 25.5              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |     |                    |      | n-Year Return Period Values of Extreme Temperature |     |            |      |            |      |            |      |       |      |
|-------------------|------|-----|----------------------------|-----|--------------------|------|----------------------------------------------------|-----|------------|------|------------|------|------------|------|-------|------|
|                   |      |     | Mean                       |     | Standard Deviation |      | n=5 years                                          |     | n=10 years |      | n=20 years |      | n=50 years |      |       |      |
| 1%                | 2.5% | 5%  | Min                        | Max | Min                | Max  | Min                                                | Max | Min        | Max  | Min        | Max  | Min        | Max  |       |      |
| (a)               | (b)  | (c) | (d)                        | (e) | (f)                | (g)  | (h)                                                | (i) | (j)        | (k)  | (l)        | (m)  | (n)        | (o)  | (p)   | (q)  |
| (4)               | 9.6  | 8.5 | 7.7                        | DB  | -20.6              | 30.4 | 4.0                                                | 1.9 | -23.5      | 31.7 | -25.8      | 32.9 | -28.1      | 33.9 | -31.0 | 35.3 |
| (5)               |      |     |                            | WB  | -20.9              | 24.2 | 3.9                                                | 0.9 | -23.8      | 24.8 | -26.1      | 25.4 | -28.3      | 25.9 | -31.1 | 26.6 |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 8.0    | -5.1  | -4.0  | 0.3   | 7.1   | 13.3  | 17.7  | 19.8  | 19.0  | 15.7  | 9.8   | 3.3   | -1.6 |
|                                                                                  | DBStd       | 10.09  | 6.32  | 6.22  | 6.30  | 5.43  | 4.72  | 3.60  | 2.81  | 2.78  | 3.84  | 4.68  | 5.20  | 5.10 |
|                                                                                  | HDD10.0     | 1946   | 469   | 392   | 307   | 119   | 21    | 1     | 0     | 0     | 3     | 63    | 208   | 362  |
|                                                                                  | HDD18.3     | 3944   | 727   | 625   | 558   | 338   | 167   | 53    | 17    | 26    | 97    | 267   | 451   | 619  |
|                                                                                  | CDD10.0     | 1221   | 1     | 1     | 8     | 32    | 123   | 232   | 305   | 279   | 173   | 57    | 8     | 1    |
|                                                                                  | CDD18.3     | 179    | 0     | 0     | 0     | 2     | 11    | 35    | 64    | 47    | 18    | 2     | 0     | 0    |
|                                                                                  | CDH23.3     | 998    | 0     | 0     | 1     | 15    | 77    | 217   | 358   | 226   | 99    | 7     | 0     | 0    |
|                                                                                  | CDH26.7     | 142    | 0     | 0     | 0     | 1     | 7     | 32    | 60    | 30    | 13    | 0     | 0     | 0    |
| Wind                                                                             | WSAvg       | 3.7    | 4.4   | 4.1   | 4.1   | 4.1   | 3.5   | 3.2   | 2.9   | 2.7   | 3.0   | 3.6   | 4.1   | 4.3  |
| Precipitation                                                                    | PrecAvg     | 1099   | 79    | 60    | 69    | 89    | 91    | 95    | 113   | 99    | 109   | 105   | 95    | 95   |
|                                                                                  | PrecMax     | 1377   | 144   | 133   | 117   | 154   | 208   | 198   | 221   | 174   | 197   | 174   | 136   | 165  |
|                                                                                  | PrecMin     | 873    | 40    | 26    | 37    | 48    | 33    | 15    | 40    | 55    | 43    | 50    | 40    | 58   |
|                                                                                  | PrecStd     | 134    | 28    | 23    | 24    | 28    | 44    | 40    | 41    | 31    | 36    | 33    | 27    | 24   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 13.7  | 13.9  | 21.3  | 25.9  | 27.6  | 29.0  | 30.6  | 28.9  | 28.0  | 24.0  | 18.2  | 14.8 |
|                                                                                  |             | MCWB   | 11.6  | 9.2   | 14.2  | 15.7  | 19.1  | 21.7  | 22.4  | 21.4  | 20.7  | 17.8  | 12.6  | 11.8 |
|                                                                                  | 2%          | DB     | 10.3  | 10.9  | 17.0  | 22.2  | 26.0  | 27.6  | 28.0  | 27.5  | 26.2  | 21.8  | 16.1  | 11.1 |
|                                                                                  |             | MCWB   | 9.1   | 7.9   | 10.6  | 13.8  | 17.9  | 20.5  | 21.4  | 20.7  | 20.0  | 16.2  | 11.5  | 9.3  |
|                                                                                  | 5%          | DB     | 7.0   | 7.6   | 12.9  | 18.8  | 23.6  | 26.2  | 27.2  | 26.1  | 23.9  | 19.0  | 13.7  | 7.9  |
|                                                                                  |             | MCWB   | 5.3   | 4.7   | 8.7   | 12.1  | 16.9  | 19.5  | 20.6  | 19.8  | 18.6  | 15.1  | 10.2  | 6.1  |
|                                                                                  | 10%         | DB     | 3.4   | 4.7   | 9.9   | 16.2  | 21.4  | 24.0  | 25.8  | 24.1  | 22.3  | 17.3  | 12.0  | 5.6  |
|                                                                                  |             | MCWB   | 1.8   | 1.9   | 6.2   | 10.2  | 15.2  | 18.2  | 19.7  | 18.9  | 17.8  | 13.7  | 9.1   | 3.5  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 11.9  | 10.7  | 14.5  | 17.0  | 20.6  | 22.8  | 24.0  | 23.2  | 22.6  | 19.9  | 14.1  | 12.4 |
|                                                                                  |             | MCDB   | 13.6  | 12.8  | 20.2  | 23.5  | 25.6  | 27.4  | 28.2  | 27.3  | 26.3  | 22.6  | 16.9  | 14.1 |
|                                                                                  | 2%          | WB     | 9.0   | 7.7   | 11.7  | 14.9  | 19.3  | 21.6  | 22.7  | 21.9  | 20.8  | 17.8  | 12.4  | 9.3  |
|                                                                                  |             | MCDB   | 9.9   | 9.5   | 14.9  | 20.1  | 24.2  | 26.2  | 26.9  | 25.3  | 24.8  | 19.8  | 14.8  | 10.4 |
|                                                                                  | 5%          | WB     | 5.4   | 4.9   | 9.1   | 13.0  | 17.8  | 20.5  | 21.6  | 21.0  | 19.7  | 15.8  | 10.7  | 6.5  |
|                                                                                  |             | MCDB   | 6.7   | 7.5   | 12.8  | 17.8  | 21.9  | 24.8  | 25.3  | 24.2  | 22.8  | 18.5  | 13.1  | 7.9  |
|                                                                                  | 10%         | WB     | 2.0   | 2.2   | 6.5   | 11.2  | 16.4  | 19.5  | 20.6  | 20.1  | 18.5  | 13.9  | 8.8   | 3.6  |
|                                                                                  |             | MCDB   | 3.5   | 4.6   | 10.3  | 15.1  | 20.2  | 23.2  | 24.0  | 23.3  | 21.1  | 16.7  | 11.4  | 5.3  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 6.9   | 7.9   | 8.3   | 10.0  | 9.8   | 9.2   | 9.1   | 8.8   | 9.1   | 8.2   | 6.9   | 5.8  |
|                                                                                  |             | MCDBR  | 9.4   | 10.8  | 12.3  | 14.0  | 11.8  | 11.0  | 10.4  | 10.4  | 11.1  | 11.1  | 10.4  | 9.0  |
|                                                                                  |             | MCWBR  | 8.6   | 8.8   | 8.2   | 8.7   | 6.5   | 5.9   | 5.3   | 5.6   | 6.0   | 6.7   | 7.7   | 7.7  |
|                                                                                  | 5% WB       | MCDBR  | 9.4   | 10.6  | 11.2  | 12.3  | 10.6  | 9.6   | 9.1   | 8.8   | 9.5   | 9.6   | 9.2   | 8.8  |
|                                                                                  |             | MCWBR  | 9.0   | 9.2   | 8.2   | 8.6   | 6.5   | 5.9   | 5.2   | 5.3   | 5.7   | 6.6   | 7.8   | 8.2  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.292  | 0.322 | 0.323 | 0.352 | 0.383 | 0.414 | 0.415 | 0.406 | 0.357 | 0.343 | 0.302 | 0.289 |      |
|                                                                                  | taud        | 2.401  | 2.299 | 2.406 | 2.386 | 2.325 | 2.262 | 2.277 | 2.291 | 2.410 | 2.449 | 2.534 | 2.470 |      |
|                                                                                  | Ebn at Noon | 855    | 876   | 924   | 917   | 891   | 861   | 855   | 852   | 875   | 846   | 844   | 830   |      |
|                                                                                  | Edh at Noon | 79     | 102   | 104   | 115   | 125   | 134   | 131   | 125   | 103   | 88    | 69    | 68    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 1.33   | 2.15  | 3.19  | 4.39  | 5.30  | 5.77  | 5.84  | 5.21  | 4.14  | 2.57  | 1.61  | 1.02  |      |
|                                                                                  | RadStd      | 0.12   | 0.24  | 0.39  | 0.47  | 0.54  | 0.46  | 0.45  | 0.30  | 0.41  | 0.26  | 0.24  | 0.13  |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|      |                         | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (48 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | +82     | N/A     |

Nomenclature: See separate page